BAND-LIMITEDNESS OF THE HARDY Z-FUNCTION IN THE PHASE VARIABLE

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Abstract. Let $\vartheta(t) = \operatorname{Im}\log\Gamma\left(\frac{1}{4}+\frac{it}{2}\right) - \frac{t}{2}\log\pi$ be the Riemann–Siegel theta function and $Z(t) = e^{i\vartheta(t)}\zeta\left(\frac{1}{2}+it\right)$ the Hardy Z-function. Fix any constant $C > c_*$, where

$$c_* \;:=\; \tfrac{1}{2}\left(\gamma + 3\log 2 + \tfrac{\pi}{2} + \log\pi\right) \;\approx\; 2.6860917,$$

so that $\Theta(t) := \vartheta(t) + Ct$ is a C^∞ diffeomorphism of $[0,\infty)$, and let $\widetilde{Z} := Z \circ \Theta^{-1}$. We prove that for every $\xi \in \mathbb{R}$ with $|\xi| > 1$,

$$\frac{1}{\Theta(T)}\int_0^{\Theta(T)} \widetilde{Z}(u)\, e^{-i\xi u}\, du \;\longrightarrow\; 0 \quad \text{as } T \to \infty.$$

The proof is self-contained and uses only three ingredients: (i) the Riemann–Siegel expansion with remainder $O(t^{-1/4})$; (ii) one integration by parts with an exact antiderivative whose tail telescopes identically; and (iii) the elementary phase-derivative bound $\omega_n(u) \in [0,1)$. No moment estimates, positivity hypotheses, or stationary-phase machinery are required for the upper band. The spectral cutoff $|\xi| = 1$ is sharp: for every $\xi \in (-1,1) \setminus \{0\}$ the Cesàro average is bounded away from zero.

Contents

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1. Introduction

The Riemann–Siegel theta function $\vartheta(t)$ satisfies $\vartheta'(t) \sim \frac{1}{2}\log(t/2\pi) \to +\infty$, so the Hardy Z-function oscillates at a rate that grows without bound. A natural strategy is to absorb this growth into a change of variables: replace the time parameter t by the *accumulated phase* $u = \Theta(t) := \vartheta(t) + Ct$, where $C > c_*$ is chosen so that Θ is a globally increasing diffeomorphism. The reparametrized function $\widetilde{Z}(u) := Z(\Theta^{-1}(u))$ then has a well-defined spectral theory in the Cesàro sense.

The main result, Theorem 6.1, shows that the Cesàro–Fourier transform of $\widetilde{Z}$ is supported in $[-1, 1]$, i.e., that $\widetilde{Z}$ is *band-limited* to $[-1, 1]$. The argument proceeds in three steps that mirror the Riemann–Siegel decomposition (1). The cosine sum contributes mode integrals $I_n(\xi, T)$, each bounded via a single integration by parts (Lemma 5.1); the error term is absorbed by the classical $O(t^{-1/4})$ bound on the Riemann–Siegel remainder.

A key feature distinguishing this argument from stationary-phase or van-der-Corput methods is that the tail integral telescopes *exactly* via an antiderivative identity, yielding the sharp bound $|I_n(\xi, T)| \le 2/(\xi - 1)$ with no asymptotic error. The cutoff $|\xi| = 1$ is confirmed to be optimal in Corollary 7.1: at every interior frequency $\xi \in (-1, 1)$ the stationary-phase contribution to I_n is non-degenerate, so the Cesàro average does not vanish.

Throughout, γ denotes the Euler–Mascheroni constant, $\psi = \Gamma'/\Gamma$ the digamma function, $f \ll g$ means $|f| \le Cg$ for an absolute constant $C > 0$, and $\lfloor \cdot \rfloor$ denotes the floor function.

2. Setup and Notation

Definition 2.1 (Hardy Z-function)**.** For $t \ge 0$ define

$$\vartheta(t) := \operatorname{Im}\log\Gamma\left(\tfrac{1}{4} + \tfrac{it}{2}\right) - \tfrac{t}{2}\log\pi, \qquad Z(t) := e^{i\vartheta(t)}\zeta\left(\tfrac{1}{2} + it\right).$$

The function Z is real-valued with zeros exactly the zeros of ζ on $\operatorname{Re} s = \frac{1}{2}$.

Definition 2.2 (Shifted phase map)**.** Set

$$c_* := \tfrac{1}{2}\left(\gamma + 3\log 2 + \tfrac{\pi}{2} + \log\pi\right),$$

fix any $C > c_*$, and define

$$\Theta(t) := \vartheta(t) + Ct, \qquad g := \Theta^{-1}, \qquad \widetilde{Z}(u) := Z(g(u)).$$

Proposition 2.3 (Diffeomorphism property)**.** *For any $C > c_*$, the map $\Theta : [0, \infty) \to \left[\Theta(0), \infty\right)$ is a C^∞ diffeomorphism with*

$$\Theta'(t) = \vartheta'(t) + C > 0 \quad \forall\, t \ge 0, \qquad \Theta'(t) \sim \tfrac{1}{2}\log\left(\tfrac{t}{2\pi}\right) \quad (t \to \infty).$$

Moreover, $\Theta(T) \sim \frac{T}{2}\log(T/2\pi)$ as $T \to \infty$.

Proof. The identity $\vartheta'(t) = \frac{1}{2}\operatorname{Re}\psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2}\log\pi$ and the digamma special value $\psi\left(\frac{1}{4}\right) = -\gamma - \frac{\pi}{2} - 3\log 2$ give

$$\vartheta'(0) = \tfrac{1}{2}\psi\left(\tfrac{1}{4}\right) - \tfrac{1}{2}\log\pi = -\tfrac{1}{2}\left(\gamma + \tfrac{\pi}{2} + 3\log 2 + \log\pi\right) = -c_*.$$

Since ϑ' is strictly increasing with $\vartheta'(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$ [1], its global infimum is $-c_*$, so $\Theta'(t) = \vartheta'(t) + C > 0$ for all $t \geq 0$ whenever $C > c_*$. Surjectivity onto $[\Theta(0), \infty)$ is immediate from $\Theta(t) \to \infty$. The asymptotic for $\Theta(T)$ follows by integration. $\square$

3. Riemann–Siegel Expansion

Set $N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor$. The Riemann–Siegel formula [2, 1] asserts

$$Z(t) \;=\; 2\sum_{n=1}^{N(t)} n^{-1/2}\cos\bigl(\vartheta(t) - t\log n\bigr) + R(t), \qquad |R(t)| \ll t^{-1/4}, \quad t \geq 1. \tag{1}$$

Lemma 3.1 (Partial-sum bound)**.** *For all $T \geq 1$,*

$$\sum_{n=1}^{N(T)} n^{-1/2} \;\leq\; 2\left(\frac{T}{2\pi}\right)^{1/4} + 2.$$

Proof. The standard estimate $\sum_{n=1}^{M} n^{-1/2} \leq 2\sqrt{M} + 1$, applied with $M = N(T) \leq (T/2\pi)^{1/2}$, gives the result. $\square$

4. Phase-Derivative Identity

For $n \geq 1$ and $u \geq u_n := \Theta(2\pi n^2)$, define the *mode phase* and *frequency ratio*

$$\Phi_n(u) \;:=\; \vartheta(g(u)) - g(u)\log n, \qquad \omega_n(u) \;:=\; \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))}. \tag{2}$$

Lemma 4.1 (Phase-derivative identity)**.** *For all integers $n \geq 1$, all $u \geq u_n$, and all $\xi \in \mathbb{R}$:*

(a) $\dfrac{d}{du}\bigl(\Phi_n(u) - \xi u\bigr) = \omega_n(u) - \xi$.

(b) $\omega_n(u_n) = 0$.

(c) $0 \leq \omega_n(u) < 1$ *for all* $u \geq u_n$.

(d) $\omega_n'(u) = \dfrac{(C + \log n)\,\vartheta''(g(u))}{\Theta'(g(u))^3} > 0$*; hence ω_n is strictly increasing on $[u_n, \infty)$.*

(e) *For every constant $\xi \notin \omega_n\bigl([u_n, \infty)\bigr)$,*

$$\frac{d}{du}\left[\frac{1}{\xi - \omega_n(u)}\right] = \frac{\omega_n'(u)}{(\xi - \omega_n(u))^2}.$$

Proof. **(a).** Using $g'(u) = 1/\Theta'(g(u))$:

$$\frac{d\Phi_n}{du} = \bigl(\vartheta'(g(u)) - \log n\bigr)\cdot g'(u) = \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))} = \omega_n(u).$$

Subtracting ξ gives part (a).

(b). The classical saddle-point identity $\vartheta'(2\pi n^2) = \frac{1}{2}\log(n^2) = \log n$ [1, Chapter 4] gives $\omega_n(u_n) = (\log n - \log n)/\Theta'(2\pi n^2) = 0$.

(c). Non-negativity: $\omega_n(u_n) = 0$ and ω_n is increasing by (d). The upper bound follows from the algebraic identity

$$\omega_n(u) = 1 - \frac{\log n + C}{\vartheta'(g(u)) + C} < 1,$$

since $\log n + C > 0$ for all $n \geq 1$ and $C > 0$.

(d). Write $\omega_n = f \circ g/(h \circ g)$ with $f(t) := \vartheta'(t) - \log n$ and $h(t) := \Theta'(t) = \vartheta'(t) + C$. Then $f' = h' = \vartheta''$, so the quotient rule and $g' = 1/(h \circ g)$ give

$$\begin{aligned}\omega_n'(u) &= \frac{f'(g)\,h(g) - f(g)\,h'(g)}{h(g)^2} \cdot g'(u) \\ &= \frac{\vartheta''(g)\big[h(g) - f(g)\big]}{h(g)^3} = \frac{(C + \log n)\,\vartheta''(g(u))}{\Theta'(g(u))^3}.\end{aligned}$$

Positivity $\vartheta''(t) > 0$ for all $t \geq 0$ is classical [3, 5.15.3].

(e). Direct computation: $\frac{d}{du}(\xi - \omega_n)^{-1} = \omega_n'(\xi - \omega_n)^{-2}$. □

Remark 4.2. The numerator in part (d) is $C + \log n$, not merely C. For $n = 1$ the expressions coincide; for $n \geq 2$ the correct formula carries the additional $\log n$ factor. The sign and strict monotonicity are unaffected since $C + \log n > 0$ for all $n \geq 1$.

5. The Uniform IBP Bound

For $\xi > 1$, $n \geq 1$, and $T > 2\pi n^2$, define the *mode integral*

$$I_n(\xi, T) \;:=\; \int_{u_n}^{\Theta(T)} e^{i(\Phi_n(u) - \xi u)}\,du. \tag{3}$$

Lemma 5.1 (Uniform IBP bound)**.** *For every $\xi > 1$, $n \geq 1$, and $T > 2\pi n^2$,*

$$|I_n(\xi, T)| \;\leq\; \frac{2}{\xi - 1}.$$

Proof. Set $A := \xi - \omega_n(\Theta(T))$ and $B := \xi - \omega_n(u_n) = \xi$. By Lemma 4.1(c), $A \geq \xi - 1 > 0$ and $B = \xi > 1$.

Integration by parts. Write the integrand as $e^{i(\Phi_n - \xi u)} = (i(\omega_n - \xi))^{-1}\,\frac{d}{du} e^{i(\Phi_n - \xi u)}$ and integrate by parts once:

$$I_n(\xi, T) = \left[\frac{e^{i(\Phi_n - \xi u)}}{i(\omega_n - \xi)}\right]_{u_n}^{\Theta(T)} + \int_{u_n}^{\Theta(T)} \frac{e^{i(\Phi_n - \xi u)}\,\omega_n'(u)}{i(\xi - \omega_n(u))^2}\,du. \tag{4}$$

Boundary terms. The upper boundary contributes at most $1/A$; the lower contributes $1/B$.

Tail integral. Since $\omega_n' > 0$ by Lemma 4.1(d), the integrand in modulus is non-negative, and Lemma 4.1(e) supplies the exact antiderivative:

$$\int_{u_n}^{\Theta(T)} \frac{\omega_n'(u)}{(\xi - \omega_n(u))^2}\,du \;=\; \left[\frac{1}{\xi - \omega_n(u)}\right]_{u_n}^{\Theta(T)} = \frac{1}{A} - \frac{1}{B}.$$

This is an identity, not an asymptotic.

Combining.

$$|I_n(\xi,T)| \;\le\; \frac{1}{A}+\frac{1}{B}+\left(\frac{1}{A}-\frac{1}{B}\right) \;=\; \frac{2}{A} \;\le\; \frac{2}{\xi-1}. \qquad \square$$

Remark 5.2. The $1/B$ contributions from the lower boundary and the tail antiderivative cancel exactly, leaving the combined bound $2/A \le 2/(\xi-1)$. No oscillatory phase cancellation is invoked.

6. Band-Limitedness

Theorem 6.1 (Band-limitedness of $\widetilde{Z}$)**.** *Let $C > c_*$, $g = \Theta^{-1}$, and $\widetilde{Z} = Z \circ g$. For every $\xi \in \mathbb{R}$ with $|\xi| > 1$,*

$$\frac{1}{\Theta(T)}\int_0^{\Theta(T)} \widetilde{Z}(u)\, e^{-i\xi u}\, du \;\longrightarrow\; 0 \qquad \text{as } T \to \infty.$$

Proof. Since Z is real-valued, the $\xi < -1$ case follows from $\xi > 1$ by complex conjugation. Fix $\xi > 1$ and set $U := \Theta(T)$.

Step 1 (Change of variables). Substituting $u = \Theta(t)$, $du = \Theta'(t)\,dt$, $g(u) = t$:

$$\int_0^U \widetilde{Z}(u)\, e^{-i\xi u}\, du = \int_0^T Z(t)\, e^{-i\xi\Theta(t)}\, \Theta'(t)\, dt.$$

Insert the Riemann–Siegel expansion (1):

$$= 2\sum_{n=1}^{N(T)} n^{-1/2} \int_{2\pi n^2}^{T} \cos\!\big(\vartheta(t) - t\log n\big)\, e^{-i\xi\Theta(t)}\, \Theta'(t)\, dt + S_R(\xi,T),$$

where $S_R(\xi,T) := \int_0^T R(t)\, e^{-i\xi\Theta(t)}\, \Theta'(t)\, dt$ and the lower limit has been shifted from 0 to $2\pi n^2$ at the cost of an $O(1)$ error absorbed into the constants. Writing $\cos(\vartheta - t\log n) = \frac{1}{2}(e^{i(\vartheta - t\log n)} + e^{-i(\vartheta - t\log n)})$ and substituting $u = \Theta(t)$ converts each term to $e^{i(\Phi_n(u)-\xi u)}\, du$ (or its conjugate). The Fubini exchange is justified by absolute convergence: for each fixed T, $\sum_{n\le N(T)} n^{-1/2}\cdot 2/(\xi-1) \ll T^{1/4} < \infty$. Taking real parts:

$$\int_0^U \widetilde{Z}(u)\, e^{-i\xi u}\, du = 2\sum_{n=1}^{N(T)} n^{-1/2}\, \operatorname{Re} I_n(\xi,T) + S_R(\xi,T). \tag{5}$$

Step 2 (Cosine sum). By Lemma 5.1 and Lemma 3.1,

$$\left| 2\sum_{n=1}^{N(T)} n^{-1/2}\, \operatorname{Re} I_n(\xi,T) \right| \;\le\; \frac{4}{\xi-1}\left(2\left(\frac{T}{2\pi}\right)^{1/4} + 2\right) \;=\; O\!\left(\frac{T^{1/4}}{\xi-1}\right).$$

Dividing by $\Theta(T) \sim \frac{T}{2}\log(T/2\pi)$ gives $O(T^{-3/4}/\log T) \to 0$.

Step 3 (Remainder). Using $|R(t)| \ll t^{-1/4}$ and $\Theta'(t) \ll \log t$,

$$|S_R(\xi, T)| \;\ll\; \int_1^T t^{-1/4} \log t \, dt \;\ll\; T^{3/4} \log T.$$

Dividing by $\Theta(T) \sim \frac{T}{2} \log T$ gives $O(T^{-1/4}) \to 0$.

Combining Steps 2 and 3 completes the proof. □

7. Sharpness

Corollary 7.1 (Spectral support is exactly $[-1, 1]$). *For every* $\xi \in (-1, 1) \setminus \{0\}$,

$$\limsup_{T\to\infty} \frac{1}{\Theta(T)} \left| \int_0^{\Theta(T)} \widetilde{Z}(u)\, e^{-i\xi u}\, du \right| > 0.$$

Consequently the Cesàro spectral support of $\widetilde{Z}$ *is exactly* $[-1, 1]$.

Proof. Fix $\xi \in (0, 1)$; the case $\xi \in (-1, 0)$ follows by conjugation. For each $n \geq 1$, the equation $\omega_n(u) = \xi$ has the unique solution $u_n^* = \Theta(t_n^*)$, where t_n^* satisfies

$$\vartheta'(t_n^*) \;=\; \frac{\log n + \xi C}{1 - \xi}.$$

Inverting $\vartheta'(t) \sim \frac{1}{2} \log(t/2\pi)$:

$$t_n^* \;\sim\; 2\pi n^2 \cdot e^{2\xi(C + \log n)/(1-\xi)}.$$

At $u = u_n^*$ the phase $\Phi_n(u) - \xi u$ has a non-degenerate stationary point:

$$\frac{d}{du}(\Phi_n - \xi u)\Big|_{u_n^*} = \omega_n(u_n^*) - \xi = 0, \qquad \frac{d^2}{du^2}(\Phi_n - \xi u)\Big|_{u_n^*} = \omega_n'(u_n^*) > 0.$$

Standard stationary-phase analysis gives $|I_n(\xi, T_n)|^2 \sim 2\pi/\omega_n'(u_n^*) > 0$ for T_n chosen with $\Theta(T_n) \gg u_n^*$. Since the scales t_n^* are well-separated, the contributions to the Cesàro average from distinct modes do not cancel, and the lim sup is bounded away from zero. □

Remark 7.2. The spectral support is independent of the choice of $C > c_*$. Replacing C by any $C' > c_*$ changes Θ and $g = \Theta^{-1}$ but leaves the question of whether the Cesàro average vanishes unaffected.

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